

Chapter 5: Conclusion and Future Work

5.1 Conclusion

This dissertation set out to determine whether zero-shot tabular foundation models can match the predictive performance of a carefully tuned gradient-boosted decision tree architecture on a real-world, multi-tier health risk classification task, and whether this capability, if present, generalizes across independently developed foundation model architectures rather than being specific to a single implementation. To answer this question, CatBoost, TabPFN, and TabFM were benchmarked on a 21,606-record hair fall risk dataset, evaluated under a consistent methodology encompassing predictive accuracy, statistical significance testing, training-data efficiency, and post-hoc explainability.

The results provide a clear and consistent answer. All three models achieved statistically indistinguishable predictive performance (accuracy range 77.88%–78.57%, McNemar's $p > 0.05$ for every pairwise comparison), despite CatBoost alone undergoing hyperparameter optimization and despite TabFM's primary evaluation using less than 3% of the training data available to the other two models. This finding held not only at full training scale but across a full scaling series (100, 500, 2,000, and full-context rows), where both foundation models consistently and substantially outperformed CatBoost at every reduced context size, with the performance gap narrowing only gradually as training volume increased. That TabFM — a model released only months before this study was conducted — matched or, at a 2,000-row context, exceeded both CatBoost and the more established TabPFN, indicates that the data-efficiency advantages of in-context tabular learning are not confined to a single architecture, but represent a genuine and reproducible property of the zero-shot tabular foundation model paradigm as a whole.

Beyond raw predictive performance, this study demonstrates that explainability need not be sacrificed when adopting these newer architectures. SHAP-based feature attribution, applied consistently across all three models despite requiring different underlying algorithms to remain computationally tractable, confirmed that each model's predictions were grounded in clinically plausible physiological and lifestyle indicators, offering a transparent basis for interpreting individual risk assessments rather than treating any of the three models as an opaque black box.

5.2 Challenges Encountered During Implementation

Beyond the primary research findings, this study's implementation phase surfaced a number of substantive technical challenges, several of which are documented in detail in Section 4.5. This section reflects on those challenges more broadly and their implications for future work involving newly released machine learning packages.

The most significant challenges centered on TabFM, reflecting its status as a package released only weeks before this study's experimental work began. Initial attempts to evaluate TabFM using its full training context resulted in prediction calls that failed to complete within reasonable timeframes, ultimately traced through systematic diagnostic testing to a combination of factors: an internal default ensemble configuration that silently multiplied computational cost, and pronounced, super-linear runtime scaling with training context size that was not evident from the package's documentation and had to be empirically measured. A related, independently confirmed upstream defect caused prediction failures specifically on single-row inputs, requiring a batch-padding workaround to be developed before SHAP-based explainability computation could proceed.

A separate class of challenges arose from environment-level incompatibilities rather than model-specific defects. TabPFN's model checkpoint loading intermittently failed on certain cloud GPU environments due to a version mismatch in PyTorch's internal checkpoint deserialization logic, requiring a targeted monkey-patch to resolve. Independently, SHAP's permutation-based explainability method proved incompatible with the numba just-in-time compilation library on both cloud platforms evaluated in this study, a failure that persisted across multiple remediation strategies before being resolved through substitution of an alternative, non-numba-dependent SHAP algorithm.

Collectively, these challenges illustrate a broader methodological consideration for research involving recently released machine learning tooling: newly published models and their supporting software packages may carry undocumented performance characteristics, default configurations, and environment-specific defects that are not apparent from published documentation or benchmark results alone, and that only become visible through direct, hands-on experimentation. Systematic diagnostic testing — isolating variables such as batch size, context size, and computational backend — proved essential to distinguishing genuine architectural limitations from incidental implementation defects, and is recommended as standard practice for any future work adopting similarly recent tooling.

5.3 Limitations of the Study

Several limitations of this study should be acknowledged. First, TabFM's primary evaluation used a substantially smaller training context (500 rows) than CatBoost and TabPFN (17,284 rows each), a direct consequence of the computational constraints documented in Section 4.5. While the supplementary matched-context analysis (Section 4.3) partially addresses this asymmetry by comparing all three models under identical, smaller training contexts, no single evaluation in this study compares all three models under fully identical, full-scale conditions, since doing so with TabFM was not computationally feasible within the available session constraints of free-tier cloud GPU platforms.

Second, this study's dataset, while constructed using domain-informed relationships grounded in established dermatological associations between clinical biomarkers, lifestyle factors, and hair loss outcomes, was not sourced from a single existing clinical registry (as disclosed in Section 2.1). The reported predictive performance should therefore be interpreted as a benchmark of model capability on a dataset with realistic, clinically-grounded feature-target relationships, rather than as a validated real-world diagnostic tool.

Third, SHAP explainability for TabPFN and TabFM was computed using a reduced training context and a sampled subset of test instances, rather than the full training set and complete test partition used for CatBoost's TreeExplainer-based analysis. While this methodological difference is disclosed and justified on computational grounds (Section 3.6, Section 4.5), it means that the two foundation models' explainability results carry somewhat greater estimation uncertainty than CatBoost's exact, full-dataset SHAP values.

Fourth, this study's per-class analysis (Section 4.1) identified the Moderate risk tier as consistently harder to classify than Low or High risk across all three models. While this study offers an explanation grounded in the ordinal structure of the target variable, it does not evaluate whether an ordinal-aware modeling approach (as opposed to standard multi-class classification) would specifically improve performance on this boundary class.

5.4 Future Work

Several directions emerge naturally from this study's findings and limitations.

Full-scale TabFM evaluation. As TabFM's supporting tooling matures and its runtime performance is optimized in future package releases, or as access to higher-memory GPU hardware becomes available, a full-context evaluation of TabFM directly comparable in scale to CatBoost's and TabPFN's would allow the three-way comparison in this study to be extended to fully matched conditions, removing the training-context asymmetry that necessitated the supplementary scaling analysis in this work.

Investigating TabFM's advantage at moderate context sizes. This study's scaling analysis (Section 4.3) found that TabFM specifically outperformed TabPFN at a 2,000-row training context, a result not observed at 100 or 500 rows. This finding was incidental to this study's primary research question and was not investigated further; a dedicated follow-up study varying context size more granularly, and examining whether this advantage is consistent across other datasets, could clarify whether this reflects a genuine architectural property of TabFM's row-and-column attention mechanism or a dataset-specific effect.

Ordinal regression approaches. Given the consistent difficulty all three models exhibited in classifying the Moderate risk tier, future work could evaluate ordinal regression formulations — which explicitly model the ordered relationship between Low, Moderate, and High risk — against the standard multi-class classification approach used in this study, to determine whether this modeling choice specifically improves performance on the boundary class.

Ensemble strategies. This study's preliminary exploration of simple equal-weight and confidence-weighted ensembling of CatBoost and TabPFN's predictions did not exceed the best individual model's performance. Future work could explore more sophisticated ensemble strategies, including stacking approaches or ensembles incorporating all three models (including TabFM), which were not evaluated in this study.

Validation on real clinical data. As access to genuine, ethically-cleared clinical hair-loss datasets becomes available, the modeling framework developed in this study could be re-evaluated on real patient data, allowing direct comparison against the domain-informed synthetic relationships used here, and providing a stronger basis for any future clinical decision-support application.

Production deployment and distillation. Given TabFM's and TabPFN's reliance on in-context learning at inference time — requiring their training data to remain available for every prediction — future work could explore model distillation

approaches, compressing either foundation model's learned behavior into a lightweight, standalone model suitable for low-latency deployment without requiring the full training context to be retained in production.

5.5 Final Remarks

This study contributes empirical evidence that zero-shot tabular foundation models, exemplified by both TabPFN and the more recently released TabFM, can match the predictive performance of a carefully tuned gradient-boosted baseline on a real-world, multi-tier health risk classification task, while offering substantial advantages in data efficiency and requiring no dataset-specific optimization effort. As tabular foundation models continue to mature — both in predictive capability and in the robustness of their supporting software tooling — they represent an increasingly practical alternative to traditional gradient boosting for structured, tabular prediction tasks, including in clinical and health-adjacent domains where transparent, explainable decision support is essential.